Part G — Scaling & systems

ml-ladder writeup · part G

2026-09-19

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Status: complete (G1–G3) · **Constraint:** a 1.1M-character corpus and a 4-core CPU, which is the binding limit on G1 and is stated wherever it bends a result
Numbers: seeds 0–2, reference seed 0 · raw values in `data/g-metrics.json`; G2's GPU arms in `data/g-gpu.json`
Reproduce: `ladder run G1, G2, G3` (needs `uv sync --extra torch`; G3 needs a Rust toolchain), or `uv run python tools/writeup_figures.py G`
Hardware: G1 and G3 on 4 CPU cores; G2's throughput and memory arms on a Kaggle Tesla T4 (sm_75), because no GPU was available locally and Colab refused a T4 all session

bridges into ignite-catalog

Every part before this one asked whether a model works. This one asks what it costs, and the three rungs are three different answers to "what does *cost* even mean here".

G1 treats cost as the independent variable: fix a compute budget, and the question is how to spend it between parameters and data. **G2** treats it as throughput and memory on a specific card, where the answer turns out to depend on the silicon more than on the code. **G3** treats it as *comprehension* cost — porting one forward pass to another framework, and finding out which of the decisions you thought you had made were actually made for you.

There is a thread running through all three, and it is not about performance. It is that **every number in this part is a property of a configuration, not of an algorithm**. "AMP is 2.5× faster" is true at width 512 and false at width 64. "bf16 needs sm_80" is true of tensor cores and false of the dtype. "Deeper is worse" was, in part F, mostly a statement about a graph-blind control. The rungs here are built so that the configuration is visible in the metric, because the alternative — quoting the headline and dropping the conditions — is how most of these numbers get repeated.

G1 — Fit your own scaling-law curve

Question. Loss falls as models and datasets grow. Does it fall in a shape that can be fitted and extrapolated — and what has to be held still for the fit to mean anything?

The math

The shape. Empirically, loss against any one of parameters, data or compute follows a power law with a floor:

$$L(x) = E + \frac{A}{x^{\alpha}}.$$

E is the irreducible loss — the entropy of the data that no model of this family can remove — and α is how fast the reducible part shrinks. The reason this shape matters is that it **extrapolates**: fitted on models you can afford, it predicts the loss of one you cannot, which is the only defensible reason to commit to a training run before seeing it work.

Where $6ND$ comes from, and what it drops. Take one pre-LN block of width d over a sequence of length L, counting a multiply-accumulate as 2 FLOPs:

term	FLOPs per sequence
QKV projection	$6Ld^2$

term	FLOPs per sequence
$QK^\top$	$2L^2d$
AV	$2L^2d$
output projection	$2Ld^2$
MLP (4× expansion)	$16Ld^2$

The d^2 terms sum to $24Ld^2 = 2L \cdot 12d^2$, and $12d^2$ is exactly the block's parameter count. **That identity is the whole derivation of "$2N$ per token"** — and it shows immediately that the rule covers only those terms. What is left is $4L^2d$ per block, or $4Ld$ per token. Taking backward as twice forward,

$$C \approx \underbrace{6ND}_{\text{the rule}} + \underbrace{12\, n_{\text{layer}} Ld\, D}_{\text{dropped}}.$$

The dropped term is $\propto L/d$ relative to the kept one, so it vanishes for wide models on short contexts and does not vanish here. At this rung's shape it is ****14.3%** of forward FLOPs**.

What must be held still. Three things, each of which is a way to accidentally measure something else:

1. *The schedule*, matched to each run's own budget. A shared schedule stops the short runs mid-decay at a high learning rate, and the resulting $L(D)$ curve is a picture of the schedule.
2. *The data*, seen once. Beyond one epoch a bigger model can memorize rather than generalize, which bends $L(N)$ for a reason that has nothing to do with scaling.
3. *What counts as N.* The convention is non-embedding parameters, and the argument is a scaling one: the body grows as d^2 while every embedding table grows as d. The positional table is the easy one to forget — 7.4% of the smallest model here — and since its share shrinks as the model widens, it inflates exactly the points that set the exponent.

The compute-optimal question. At a fixed budget C, sweeping N traces an IsoFLOP curve whose minimum is the best split between model size and tokens. Chinchilla found $N^* \propto C^{0.5}$ and $D^* \propto C^{0.5}$ — roughly 20 tokens per parameter. The two exponents sum to 1 exactly when FLOPs per token are proportional to N; the attention term above breaks that, so the sum is a measurement rather than an identity.

What I built. The char transformer from rung C3, swept three ways: five widths at a fixed 1M-token budget, four token budgets at a fixed width, and three IsoFLOP contours over six widths. A fourth arm tests Kaplan's claim that loss depends on parameter *count* and barely on their arrangement, by comparing 96×2 against 48×8 — the same $12d^2 n_{\text{layer}}$, four times the depth at half the width. The power-law fit is written out rather than imported, profiling over the floor E on a grid because for a fixed E the problem is an ordinary log-log regression.

Results. Medians over seeds 0–2.

$L(N)$ at a fixed 1M-token budget, and $L(D)$ at width 64:

width	non-embedding N	loss	tokens	epochs	loss
32	25,472	2.435	124,928	0.12	2.703
48	56,640	2.347	249,856	0.25	2.553
64	100,096	2.255	499,712	0.50	2.420

width	non-embedding N	loss	tokens	epochs	loss
96	223,872	2.095	999,424	1.00	2.255
128	396,800	2.004			

$$L(N) = E + \frac{A}{N^{0.074}} \quad (R^2 = 0.978), \qquad L(D) = E + \frac{B}{D^{0.084}} \quad (R^2 = 0.995).$$

Both fit well and **neither pins its floor.** The argmin lands at $E = 0$ in both cases, but floors anywhere from 0 to **0.71** (for N) and 0 to **0.76** (for D) fit essentially as well, each bringing its own exponent — 0.074 to 0.110, and 0.084 to 0.117. Over one decade of x these are the honest error bars, and they are wider than the difference between many published exponents.

The compute accounting. At width 64 with a 64-token context, attention is **14.3%** of forward FLOPs, so $6ND$ undercounts the real cost by **20.9%** (`flops/compute_over_rule` 1.209, identical on every seed because it is arithmetic).

IsoFLOP.

C	best width	N^*	D^*	tokens/param	bracketed?
3×10^{11}	16	6,592	4.43M	672	no — at the grid edge
9×10^{11}	24	14,496	6.97M	481	yes
2.7×10^{12}	64	100,096	3.79M	37.8	yes

$N^* \propto C^{1.238}$ and $D^* \propto C^{-0.072}$, against Chinchilla's 0.5 and 0.5. **That is not a measurement of the Chinchilla exponent and the rung says so** (`isoflop/exponent_is_trustworthy` $= 0$): only two of three contours bracket their own minimum, and D^* is flat because the corpus has run out — the largest contour point repeats the training split **39.7 times**.

Shape. 96×2 (223,872 parameters) scores **2.095**; 48×8 (226,272, 1.1% more) scores **2.221**. A gap of **0.126** at matched N, which is `shape/gap_in_law_terms` **1.18** — larger than one step of the width sweep that spans a 15× parameter range.

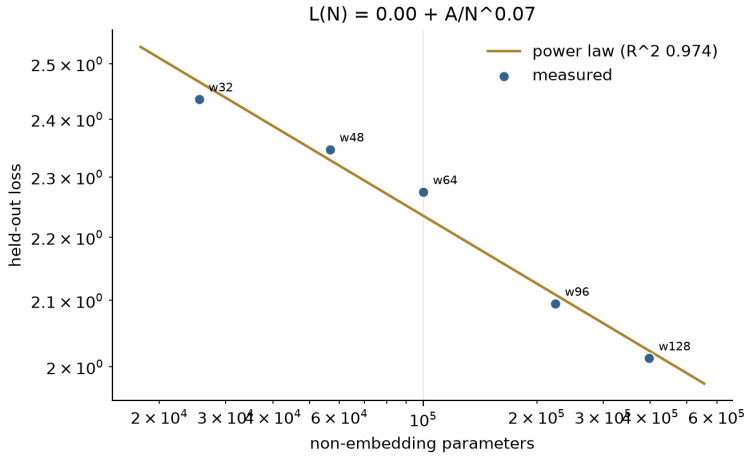


Figure 1: G1 loss vs parameters

What surprised me. "Only N matters" failed, and failed by more than the axis it was competing with. Rearranging 224k parameters from two wide layers into eight narrow ones costs 0.126

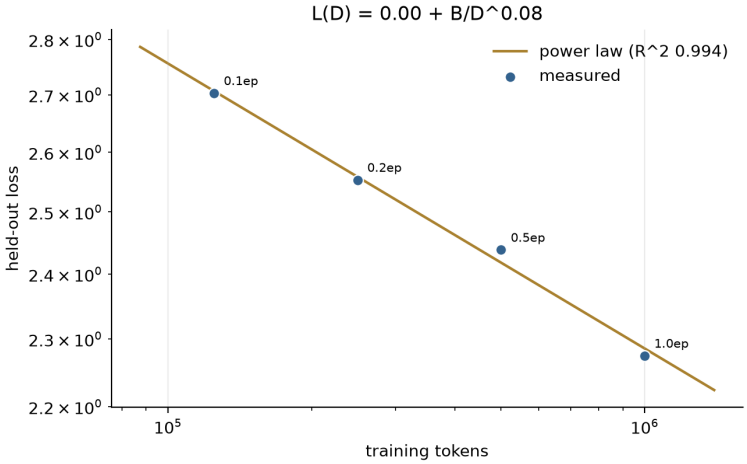


Figure 2: G1 loss vs tokens

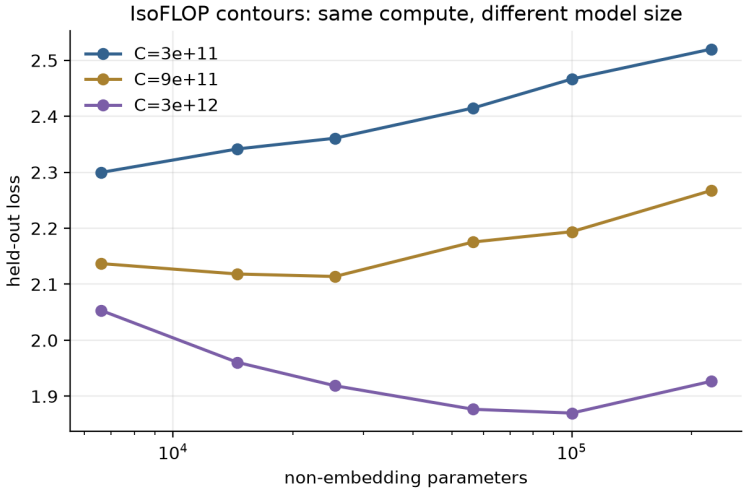


Figure 3: G1 IsoFLOP

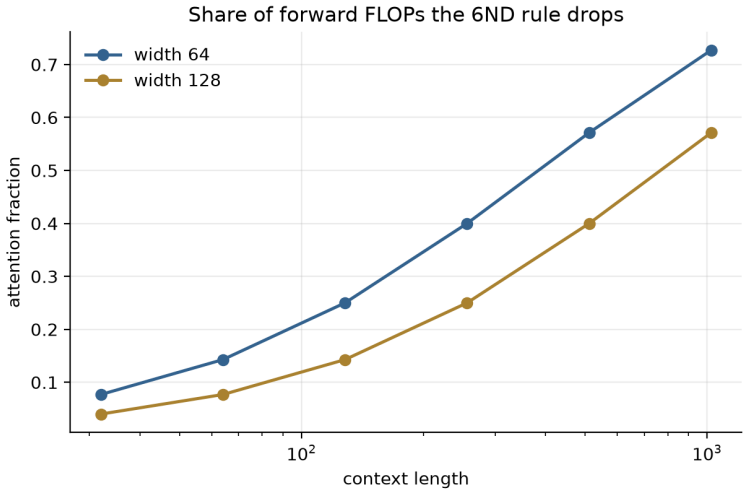


Figure 4: G1 attention share

Where 6ND comes from, and the one term it omits

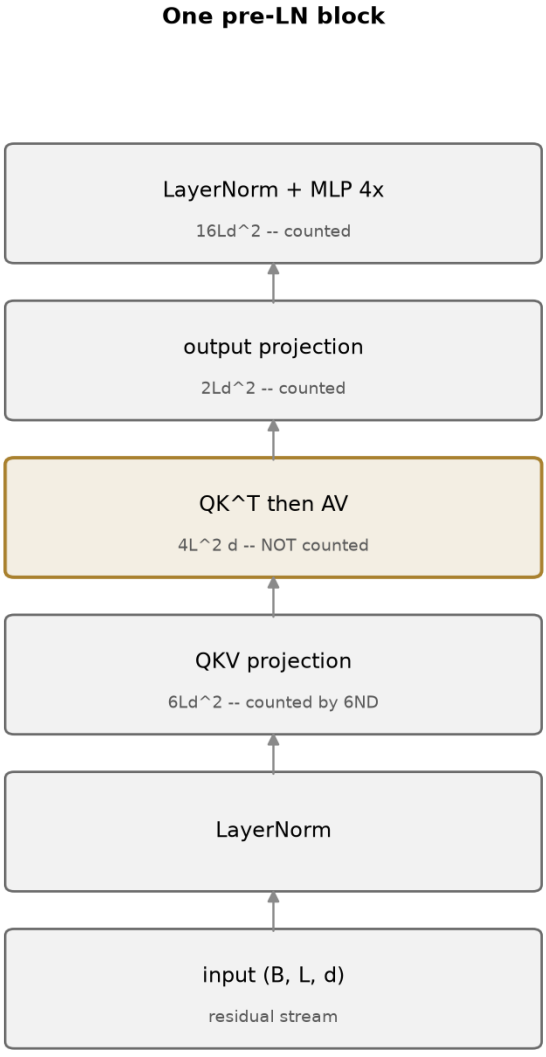


Figure 5: G1 FLOP accounting

of loss — more than the average gap between adjacent points of a width sweep covering 25k to 397k parameters. Kaplan's claim holds within a band of aspect ratios and 48×8 is outside it; at width 48 the model is deep and thin enough that depth stops being free. Part F's depth sweep found the same thing from the other side, where a deep narrow MLP was the arm that collapsed.

The floor is the parameter everyone quotes and the one the data does not support. The fitted E is 0 in both laws and the residual barely moves across floors up to 0.71 — so a paper reporting a confident irreducible loss from a one-decade sweep is reporting the argmin of a nearly flat objective. Writing the fit out instead of importing one is what made this visible: `curve_fit` returns three numbers and a covariance nobody reads, while a grid over E shows the flatness directly. The exponent inherits the same width: quoting $\alpha = 0.074$ without saying it could be 0.110 would be quoting a choice.

The compute-optimal sweep measured the corpus, not the frontier. D^* came out essentially constant — $C^{-0.072}$ — with every extra FLOP going into parameters. That is what running out of data looks like: at 1M unique tokens the optimum sits at 4 epochs and cannot move right, so the frontier bends into the N axis. The guard that made this legible was cheap — flag whether each contour's minimum is interior — and without it I would have reported $N^* \propto C^{1.24}$ as a finding rather than as a symptom.

And the smallest thing on the list mattered most. Excluding the positional table from N moves the smallest model by 7.4% and the largest by 0.5%. That is a systematic tilt applied precisely to the points that set the slope — the kind of error that does not look like an error, because every individual number stays plausible.

Open threads. Every limitation here is the corpus. A scaling law wants at least two decades of x and data it never has to repeat; 1.1M characters gives about one decade and runs out. The obvious next step is a tokenized corpus large enough that D^* is free to move, at which point the Chinchilla exponent becomes measurable rather than merely bounded.

Reading: Kaplan et al., *Scaling Laws for Neural Language Models* (2020); Hoffmann et al., *Training Compute-Optimal Large Language Models* (2022).

G2 — Mixed-precision + gradient-accumulation training, profiled

Question. Two things everybody turns on. What do they actually buy, what do they cost, and how would you know if one of them had silently broken your run?

The math

Two 16-bit formats, and the entire difference is the exponent field.

$$\text{fp32} : 1 + 8 + 23, \qquad \text{fp16} : 1 + 5 + 10, \qquad \text{bf16} : 1 + 8 + 7.$$

fp16 keeps mantissa bits and loses range: its smallest normal is 6.1×10^{-5}, and gradients live below it, so they flush to zero. **bf16 has fp32's exponent field exactly** — same smallest normal, 1.18×10^{-38} — and pays 8× coarser resolution for it. That is why `GradScaler` exists for one format and not the other: scaling the loss by a large constant moves gradients back into fp16's representable range, and the scaler's job is to find the largest constant that does not overflow, backing off whenever it does.

Gradients are exactly the wrong thing to store in the format that traded range for precision — which is the one-sentence argument for bf16, and the reason Ampere added it.

Gradient accumulation is a statement about the loss, not the gradients. For micro-batches $B_1, \dots, B_k$ partitioning an effective batch B,

$$\nabla L_B = \frac{1}{|B|} \sum_{i \in B} \nabla \ell_i = \frac{1}{k} \sum_{j=1}^{k} \nabla L_{B_j},$$

since each L_{B_j} is already a mean over its own micro-batch. The $1/k$ is not optional. Omit it and the summed gradient is k times the large-batch one, so the run trains at k times the configured learning rate — a bug that produces plausible-looking divergence rather than an error.

What I built. Five arms on a 384-wide, 4-layer transformer: fp32, fp16 with a scaler, bf16, fp16 *without* a scaler, and a sweep of accumulation counts at a fixed effective batch. Throughput is measured after a warmup with an explicit device synchronize, because CUDA's asynchrony otherwise makes the loop measure how fast Python can enqueue work. A separate sweep varies model width, because AMP's speedup comes from tensor cores and therefore from matrix shapes.

Results. On a Kaggle Tesla T4 (sm_75), single run, seed 0 — `data/g-gpu.json`. The CPU sweep in `data/g-metrics.json` covers only the arms that need no GPU, and says so.

arm	throughput	peak memory	final loss
fp32	66.6k tok/s	578.2 MB	2.5450
fp16	**168.9k (2.54×)**	441.4 MB (0.76×)	2.5450
bf16	**51.2k (0.77×)**	442.6 MB	2.5450

AMP's speedup against width, which is the claim the headline hides:

width	64	128	256	512
fp16 / fp32	1.25×	1.53×	2.31×	**2.99×**

Gradient accumulation at a fixed effective batch of 32:

accumulation	micro-batch	throughput	peak memory
1×	32	166.9k tok/s	441.4 MB
2×	16	151.9k	314.2 MB
4×	8	89.3k	231.4 MB
8×	4	45.6k	**184.7 MB**

8× buys **2.39× less peak memory for 3.66× the time**, with a loss gap of 5×10^{-7} against the single-step run — the same optimization, fed differently. And summing micro-batch gradients instead of averaging them gives a norm exactly **4.00×** larger at 4× accumulation, reproduced on CPU and GPU alike.

What surprised me. bf16 ran, and ran slower than the format it replaces. I had written that bf16 "needs sm_80, so on a Turing T4 there is no bf16 path at all, and the arm is skipped". Wrong:

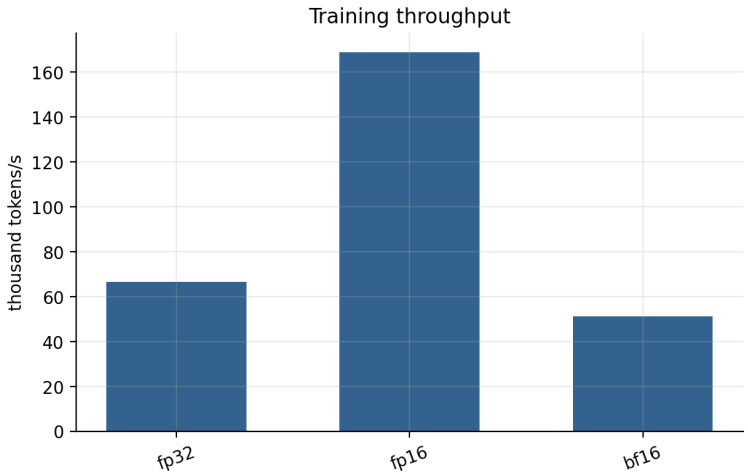


Figure 6: G2 throughput

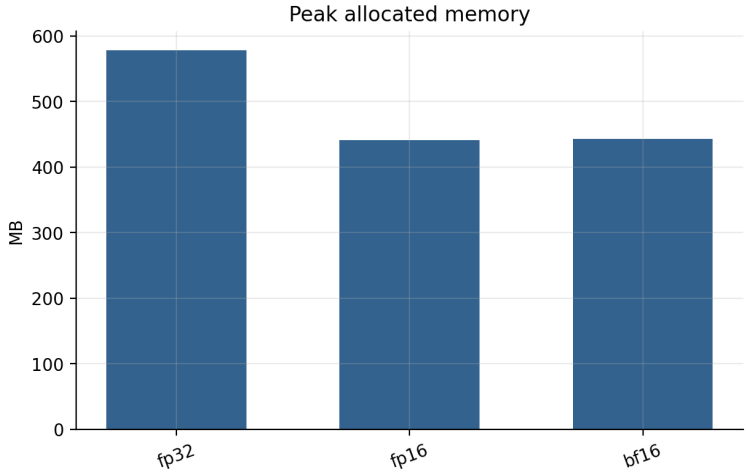


Figure 7: G2 memory

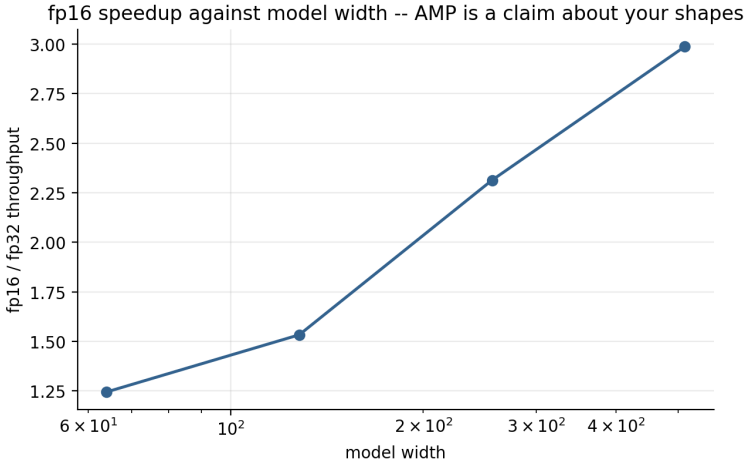


Figure 8: G2 AMP by width

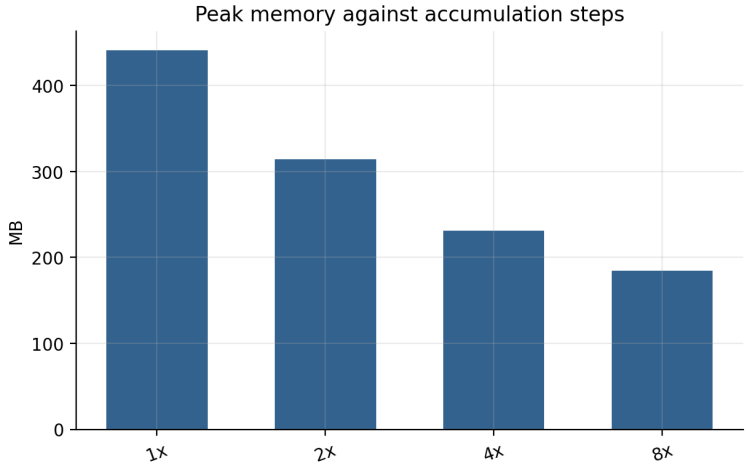


Figure 9: G2 accumulation memory

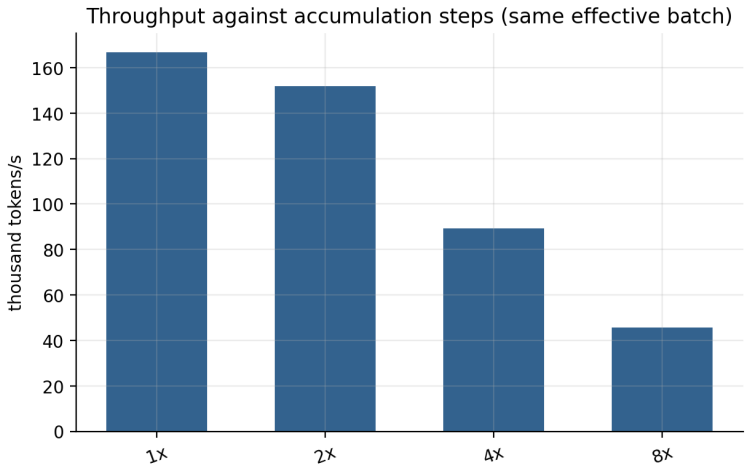


Figure 10: G2 accumulation throughput

`torch.cuda.is_bf16_supported()` returns `True` on a T4, the arm runs fine, and it lands at **0.77× fp32**. A raw 4096-square matmul on the same card, measured twice on separate sessions, says why:

dtype	TFLOP/s	
fp32	3.83 / 4.34	
fp16	**23.07 / 25.67**	tensor cores
bf16	**2.07 / 2.42**	emulated

bf16 is **11× slower than fp16 and slower than fp32 itself** here. "Supported" and "accelerated" are different questions and only throughput answers the second, so the rung now records `precision/bf16/accelerated` rather than trusting the capability flag. The same confusion is worth separating from the candle result recorded in part E: *that* failure — `DriverError(CUDA_ERROR_NOT_FOUND, "named symbol not found")` — is a **build** with no bf16 symbol, compiled under `CUDA_COMPUTE_CAP=75`. A missing kernel and a slow kernel are different problems with the same cause sitting behind them.

The scaler did nothing, and that is the result. fp16 without a `GradScaler` produced **0 non-finite steps** and a final loss of 2.5476 against 2.5450 with one. The underflow mechanism is real and the

arithmetic is right; it simply does not bite on a 384-wide model over 40 steps. Reporting the scaler as having saved the run would have been repeating the textbook instead of the measurement.

AMP's speedup is 1.25× or 2.99× depending on a number nobody quotes with it. At width 64 the matrices are too small to keep tensor cores busy and the overhead of casting eats most of the gain. "Mixed precision gives ~2×" is true of the shapes people benchmark and false of half the models in this repo.

Accumulation's loss gap is 5×10^{-7}, which is the point. It is supposed to be a pure memory-for-time trade with the optimization untouched, and at that gap it demonstrably is. The metric worth having is the one that could have caught the $1/k$ bug, and it does: 4.00× exactly, on both devices.

Open threads. Everything here is one card. The bf16 result would invert on an A100 and the accumulation trade would shift with activation checkpointing, which is the other half of the memory story and is not implemented. The obvious next measurement is `torch.compile`, which changes the shape-dependence of the AMP result by fusing away the small-kernel overhead that dominates at width 64.

Reading: Micikevicius et al., *Mixed Precision Training* (2018); the PyTorch AMP examples in this part's lab reference.

G3 — Rust/Candle port of one transformer-block forward pass

Question. Does writing a forward pass in another framework teach you anything the original did not — and can "the port is correct" be made into a number?

The math

There is no new mathematics in a port, which is the point: the function is fixed and the only question is whether two implementations compute it. That makes the measurable quantity a distance between implementations,

$$\varepsilon_{\text{rel}} = \frac{\|F_{\text{torch}}(x) - F_{\text{candle}}(x)\|}{\|F_{\text{torch}}(x)\|},$$

and the useful thing about it is that its *scale* diagnoses the cause. Around 10^{-7} in float32 is accumulation order — the same arithmetic in a different sequence. Around 10^{-5} to 10^{-3} is a different function: an approximation swapped for an exact form, a wrong epsilon, a mask applied in the wrong place. Several orders larger is a shape or orientation error.

The one function with two definitions in this block is GELU:

$$\text{exact: } x\,\Phi(x), \qquad \text{tanh: } \tfrac{x}{2}\left(1 + \tanh\left[\sqrt{\tfrac{2}{\pi}}\,(x + 0.044715x^3)\right]\right).$$

PyTorch's `nn.GELU()` is the first; Candle's `gelu()` is the second and `gelu_erf()` is the first. They agree to about 10^{-3} on ordinary activations — far above float32 noise, far below anything that looks like a bug.

What I built. The pre-LN block from C3, rewritten against Candle 0.11 in `rust/crates/ignite-port`, with a driver that exports the weights to `.npz`, runs both sides and diffs them at batch 1 and batch 4. Both GELU variants are run, so the cost of the wrong choice is a measurement rather than a warning.

Results. Medians over seeds 0–2.

	relative error	max |diff|
exact GELU, batch 1	8.4×10^{-8}	4.8×10^{-7}
exact GELU, batch 4	8.2×10^{-8}	4.8×10^{-7}
tanh GELU, batch 1	3.6×10^{-5}	1.4×10^{-4}
tanh GELU, batch 4	3.9×10^{-5}	1.4×10^{-4}

With the matching GELU the port agrees to **float32 round-off**. With Candle's default it is **470× worse** — from swapping one function, with nothing raised and nothing crashed.

Speed, on an idle machine and a loaded one:

torch threads	idle	4 competing processes
1	0.465 ms	0.959 ms
2	0.429 ms	0.936 ms
4	0.480 ms	**90.6 ms**
spread	**1.14×**	**94.6×**

Contention costs the four-thread pool **191×** (range 160–222) and the single thread **2.06×**.

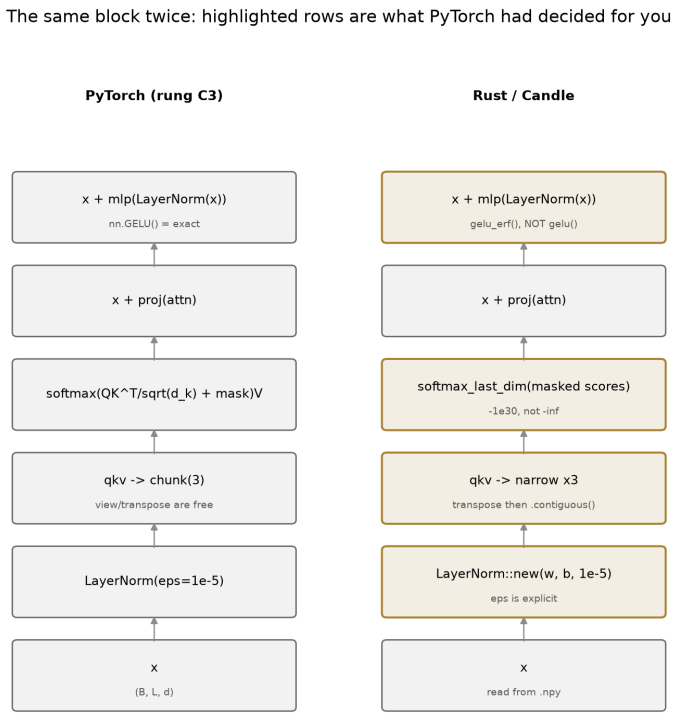


Figure 11: G3 port

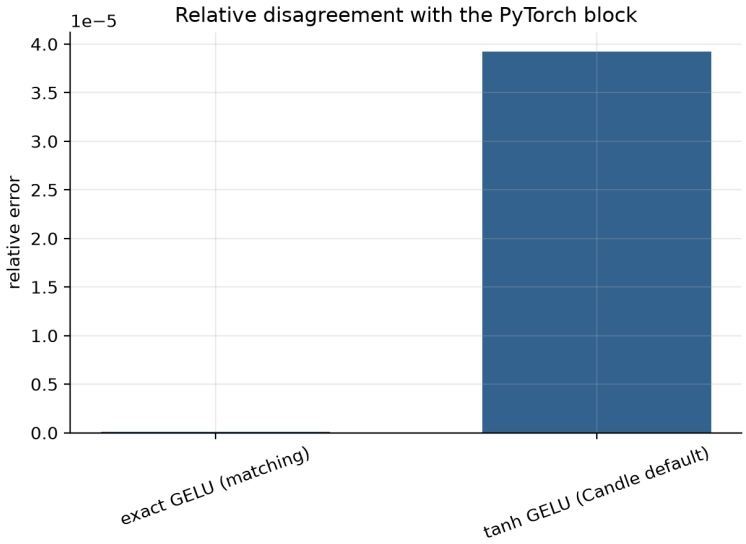


Figure 12: G3 agreement

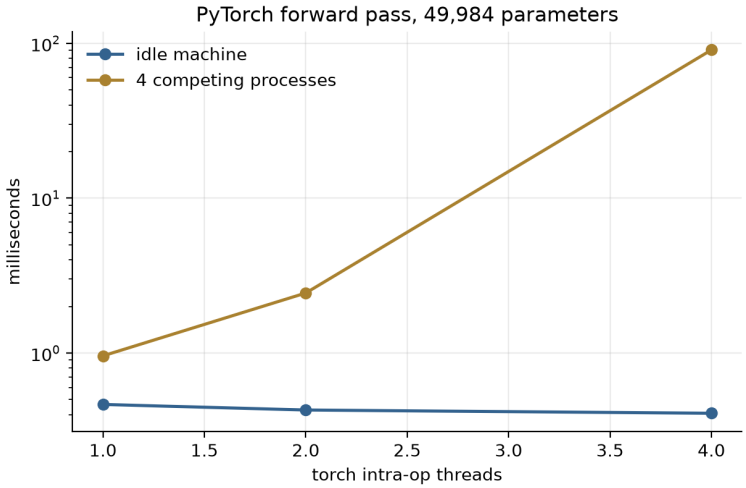


Figure 13: G3 threads

What surprised me. The port was right the first time, which was not the plan. It compiled and matched to 10^{-7} on the first run, and the value of the exercise turned out to be in the three lines where Candle refuses to decide for you: `LayerNorm::new(w, b, 1e-5)` makes the epsilon explicit, `transpose` returns a strided view so `matmul` needs an explicit `.contiguous()`, and `gelu()` is not `gelu_erf()`. The last of those is the one with no safety net — it is a different function, not a different precision, and the only thing that catches it is a diff against a reference.

The contiguity one is why the rung tests at batch 4. It does not fail at batch 1, and batch 1 is the only shape a single-sequence demo ever runs. Earlier in this same session that exact bug was found in two places in the candle fork's vision path, both of which the executor never hit because it runs at batch 1.

This rung's speed section measured the wrong thing twice. The first version compared PyTorch against the ported binary and made Rust look 10× faster; it was timing process startup and an `.npz` parse. The second fixed that, measured PyTorch at 1/2/4 threads, found 84 ms at four threads and reported *192×, from a setting* — but that run shared a 4-core box with this part's own sweep. On an idle machine the spread is **1.14×**.

What survives is better than either mistake: the cost is not the thread pool, it is the **interaction** between a spin-waiting pool and a machine that has no cores to give. Single-threaded, contention costs 2.06×; at four threads it costs 191×. So `torch.set_num_threads(1)` is worth nothing on an idle box and is the difference between 0.96 ms and 91 ms on a busy one. The rung now creates the contention itself, so the number is a measurement rather than an accident of what else was running — which is the general lesson, and one this part had to learn twice.

Open threads. One block is not a model: the port has no KV cache, no sampling loop and no quantization, which is where the [[ignite-catalog]] executors actually live. Porting the *decode* path is the rung after this one, and the interesting number there is tokens per second rather than a relative error.

Reading: the Candle source for `candle-nn`'s `LayerNorm` and `ops::softmax`; Hendrycks & Gimpel, *Gaussian Error Linear Units* (2016) for the two forms.

What part G settled

1. **$6ND$ is a rule of thumb with a named error term, and the term is 20.9% here.** The d^2 terms of a block are exactly $2L$ times its parameter count, which is where the rule comes from and also why it covers nothing else. What it drops is $4Ld$ per token — 14.3% of forward FLOPs at width 64 on a 64-token context, growing with L/d. Fine for GPT-3-shaped models, not fine for these, and either way it is a derivation rather than a constant.
2. **A power law fitted over one decade does not pin its floor, and the floor is the number people quote.** Both fits here land at $E = 0$ with $R^2 > 0.97$, and floors up to 0.71 and 0.76 fit essentially as well, carrying exponents from 0.074 to 0.110 and 0.084 to 0.117. The honest report is the band.
3. **"Loss depends on N, not on shape" is false outside a band of aspect ratios.** At matched parameter count, 48×8 loses to 96×2 by 0.126 — more than one step of a width sweep spanning 15× in N.
4. **The compute-optimal frontier measured the corpus.** $D^* \propto C^{-0.072}$, flat, because 1M unique tokens runs out at ~4 epochs and every further FLOP goes into parameters. Two of three contours bracket their minimum; the rung reports that rather than the exponent,

because an exponent fitted through edge-pinned points is a measurement of the grid.

5. **"Supported" and "accelerated" are different questions.** bf16 runs on a T4 — `is_bf16_supported()` says `True` — at 0.77× fp32 in training and 2.1–2.4 TFLOP/s against fp16's 23–26 on a raw matmul. Only throughput separates them. Separately, the candle fork's `DriverError(CUDA_ERROR_NOT_FOUND)` on the same card is a *third* thing: a build compiled for sm_75 with no bf16 symbol at all. One card, three different answers to "does bf16 work", and the correct one depends on which layer you ask.
6. **Every headline in this part is conditional on a number the headline omits.** AMP is 1.25× at width 64 and 2.99× at 512. Accumulation is free (loss gap 5×10^{-7}) and costs 3.66× the time. `set_num_threads(1)` is worth nothing idle and 191× under contention. The pattern is the part's actual content: these are properties of configurations, and quoting them without the configuration is how they turn into folklore.
7. **A port is a comprehension test with a pass mark.** 8×10^{-8} relative says the two implementations compute the same function; 4×10^{-5} says one of them computes a different one. The scale diagnoses the cause, which is why the number is worth more than a passing test — and why swapping `gelu_erf` for `gelu`, which raises nothing, is caught in one line.
8. **I measured the harness instead of the code twice in one rung.** First comparing a whole Rust process against an in-process PyTorch call, then reporting a 192× thread spread that was contention with this part's own sweep. Both produced plausible numbers, both were reported before being checked, and both were caught by re-running under changed conditions. That is now nine occasions in this repo where a number that looked like a result was an artifact of how it was computed — and the only method that has reliably caught them is adding the control that could embarrass it.

Cost of the part: ~1,750 lines including tests, 12 test cases, and a three-seed CPU sweep in about 44 minutes (G1 ~838 s per seed, G2 and G3 seconds each), plus a Kaggle T4 kernel for G2's GPU arms. G1 dominates because its IsoFLOP contours train 18 models per seed.

Figures regenerate with `uv run python tools/writeup_figures.py G`; the numbers they are quoted against live in `data/g-metrics.json` and, for the GPU arms, `data/g-gpu.json`. The part README is the short version of this document.